MARITIME LOGISTICS INTELLIGENCE

GLOBAL SUPPLY CHAIN DISRUPTION

2023–2025 Red Sea Maritime Crisis Analysis

SQL-Based Analysis of Route Diversion, Freight Rates, Port Delays, Trade Volume, Trade Volume, Emissions & Supply Chain Vulnerability

SQL | Supply Chain Analytics | Maritime Logistics | Business Intelligence

GLOBAL MARITIME ROUTES
2023–2025 RED SEA DISRUPTION ANALYSIS

Key Corridors Impacted by Red Sea Disruption

ROUTE LEGEND

- Asia–Europe via Suez (Traditional Route)
- Asia–Europe via Cape of Good Hope (Alternative Route)
- Middle East–Europe Oil Route (Energy Corridor)
- Mediterranean Feeder (Regional / Short Sea)

- Major Port
- Key Transshipment / Hub Port
- Red Sea Disruption Area

KEY PORTS IN DATASET

- Port Said (Egypt)
- Jeddah (Saudi Arabia)
- Dubai / Jebel Ali (UAE)
- Rotterdam (Netherlands)
- Hamburg (Germany)
- Antwerp (Belgium)
- Piraeus (Greece)
- Aigiecs (Algeria)
- Novorossiysk (Russia)
- Ceyhan (Turkey)
- Basra (Iraq)
- Singapore (Singapore)
- Shanghai (China)
- Ningbo (China)
- Shenzhen (China)
- Cape Town (South Africa)
- Aden (Yemen)

ROUTE SUMMARY COMPARISON

Route	Approx. Distance* (nm)	Key Characteristics	Status (2023–2025)
Asia–Europe via Suez	~8,500	Shortest route via Suez Canal & Red Sea	Disrupted / High Risk
Asia–Europe via Cape of Good Hope	~11,800	Longer but safer alternative around Africa	Active Alternative
Middle East–Europe Oil Route	~6,200	Energy shipments from Gulf to Europe	Partially Disrupted
Mediterranean Feeder	~1,500	Short-sea regional connections	Operational

*Distances are approximate and based on standard_distance_nm from dataset.

KEY INSIGHTS

- Disruption in the Red Sea forces rerouting around Africa, adding distance, time & cost.
- Suez-linked routes show highest exposure to delay, cost pressure & emissions.
- Longer routes via Cape of Good Hope increase fuel consumption & GHG emissions.
- Supply chain resilience depends on route diversification & real-time risk monitoring.

THE BUSINESS PROBLEM

WHY THE RED SEA DISRUPTION MATTERED

“Rerouting away from the Suez Canal reshaped both the cost structure and the operational flow of global maritime shipping.”

ROUTE COMPARISON

ASIA → SUEZ CANAL → EUROPE

Shorter, standard corridor — the pre-crisis default

ASIA → CAPE OF GOOD HOPE → EUROPE

Longer diversion around Africa — the crisis alternative

UNCTAD CONTEXT · Suez transits fell 40%+ from peak by Jan 2024 · Container tonnage Container tonnage via Suez fell 82% by early Feb 2024 · Cape of Good Hope tonnage tonnage rose 60% Dec 2023–Feb 2024

FOUR CASCADING IMPACTS

- LONGER ROUTES → HIGHER OPERATIONAL EXPOSURE
- FREIGHT PRESSURE → HIGHER COST RISK
- DELAYS → SUPPLY CHAIN UNCERTAINTY
- REROUTING → HIGHER EMISSIONS

Source: UNCTAD, Navigating Troubled Waters, February 2024

DATA + SQL ANALYTICAL FRAMEWORK

FROM OPERATIONAL DATA TO BUSINESS INSIGHT

ROUTES_MASTER
key: route_id
Distance

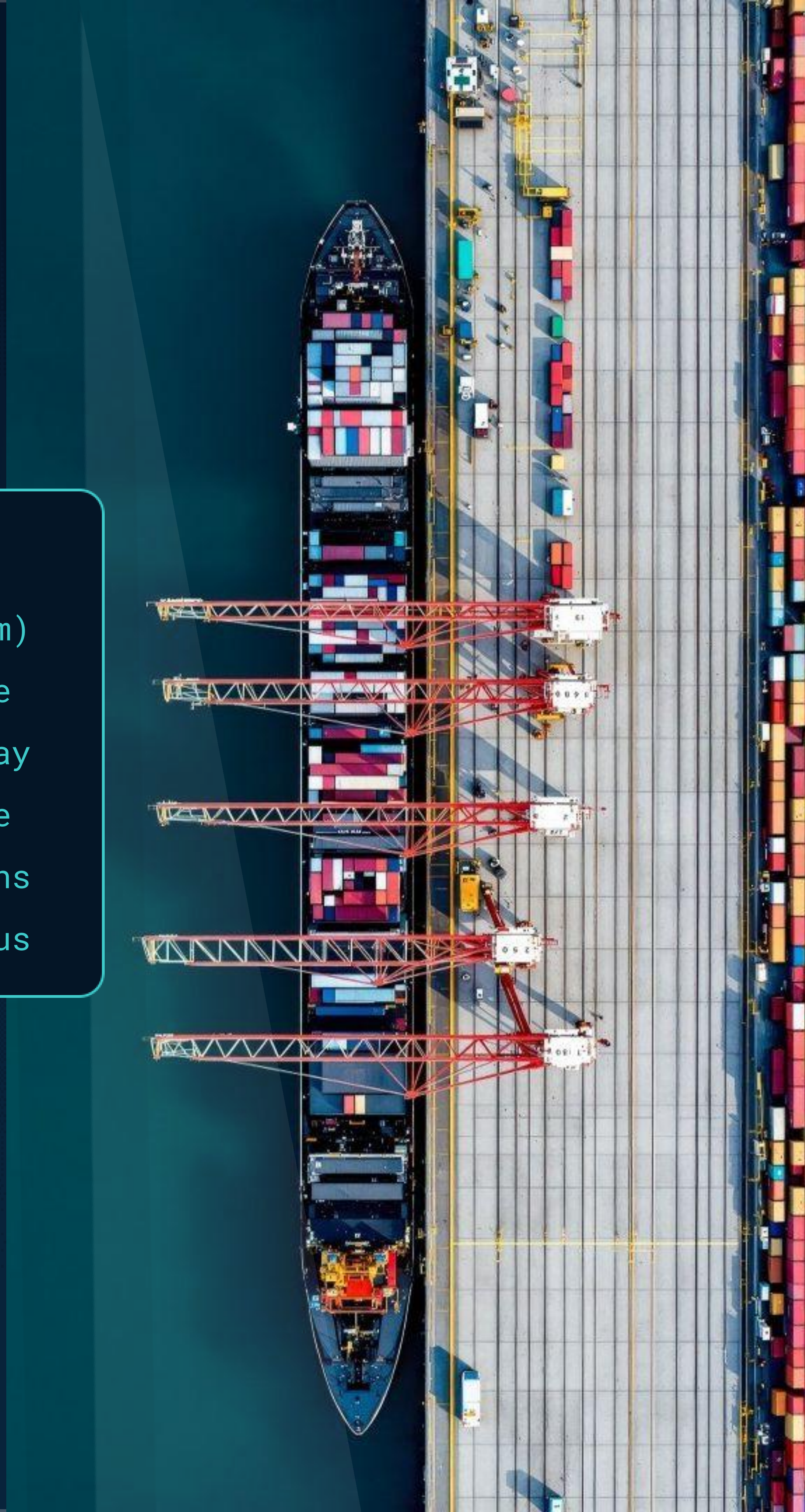
PORTS_CONGESTION_MASTER
key: port_id
Transit Delay

VESSELS_MASTER
key: vessel_type_id
Freight Rate

CRISIS_METRICS
10,000 records · central fact table

FACT FIELDS:

- Distance (nm)
- Freight Rate
- Transit Delay
- Trade Volume
- GHG Emissions
- Period Status



SQL CAPABILITIES

JOIN · GROUP BY · AGGREGATION · CTE · CASE LOGIC · WINDOW FUNCTION / RANK

Five business questions translate operational disruption into measurable supply-chain risk.

ANALYTICAL ROADMAP

FIVE BUSINESS QUESTIONS

Five business questions convert operational disruption into a measurable business evidence using SQL.

01
ROUTE DIVERSION DISTANCE

How much additional distance is created by avoiding the Suez Canal? Canal?

02
FREIGHT RATES BY VESSEL

How did freight rates vary vary across different vessel vessel types and phases?

03
PORT TRANSIT DELAYS

Which ports experienced experienced the highest transit delays during the crisis?

04
TRADE VOLUME & EMISSIONS

How did trade volume and and GHG emissions change change across disruption disruption phases?

05
CORRIDOR VULNERABILITY

Which trade corridor appears the most operationally vulnerable vulnerable overall?

BUSINESS QUESTION 01

HOW MUCH ADDITIONAL DISTANCE IS CREATED BY AVOIDING SUEZ?

SQL

```
SELECT
    r.route_name,
    r.standard_distance_nm,
    r.diverted_distance_nm,
    (r.diverted_distance_nm - r.standard_distance_nm)
AS extra_distance_nm,
    ROUND(AVG(c.distance_nm), 2) AS
avg_actual_distance_nm,
    COUNT(c.record_id) AS total_trips
FROM crisis_metrics c
JOIN routes_master r
    ON c.route_id = r.route_id
GROUP BY
    r.route_name,
    r.standard_distance_nm,
    r.diverted_distance_nm
ORDER BY extra_distance_nm DESC;
```

KPI · LARGEST DIVERSION GAP

+3,300 NM

Extra Distance Created by Route Diversion

INSIGHT

Asia–Europe via Suez has the largest modeled diversion gap at **3,300 nautical miles**, making it the making it the corridor with the greatest additional routing burden in the analytical dataset. dataset.

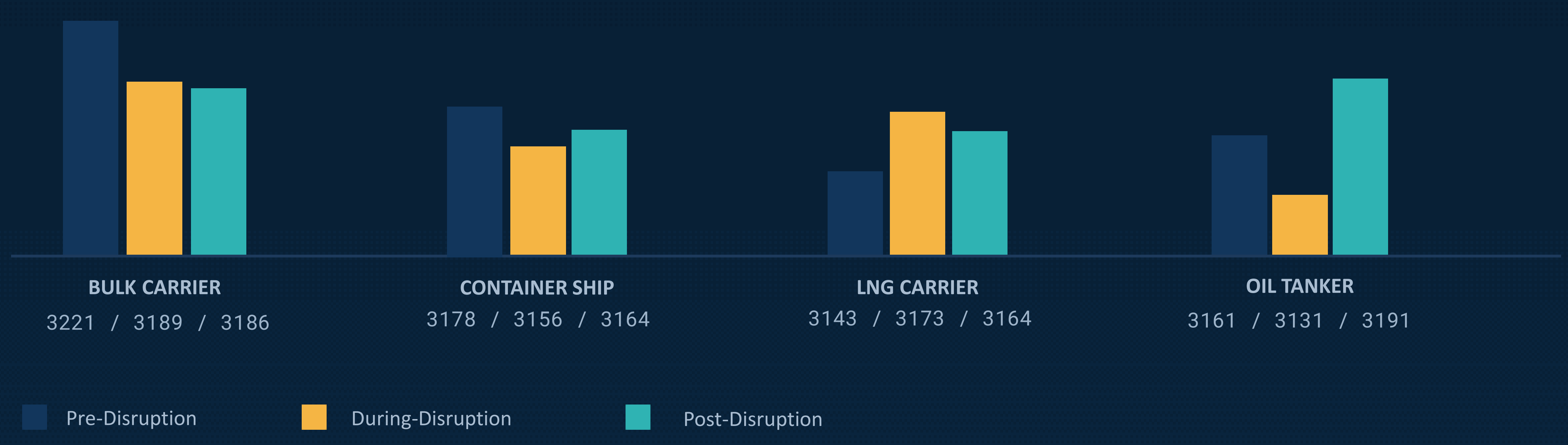
BUSINESS QUESTION 02

HOW DID FREIGHT RATES VARY ACROSS VESSEL TYPES?

SQL

```
SELECT
    v.vessel_type,
    c.period_status,
    COUNT(c.record_id) AS total_shipments,
    ROUND(AVG(c.freight_rate_usd), 2) AS
avg_freight_rate_usd,
    ROUND(MAX(c.freight_rate_usd), 2) AS
max_freight_rate_usd,
    ROUND(MIN(c.freight_rate_usd), 2) AS
min_freight_rate_usd
FROM crisis_metrics c
JOIN vessels_master v
    ON c.vessel_type_id = v.vessel_type_id
GROUP BY
    v.vessel_type,
    c.period_status
ORDER BY
    v.vessel_type,
    c.period_status;
```

Average Freight Rate (USD) by Vessel Type & Phase



INSIGHT

Freight-rate movements varied by vessel type rather than following a single disruption pattern, with oil tankers peaking post-disruption and LNG carriers during the disruption carriers during the disruption phase.

BUSINESS QUESTION 03

WHICH PORTS EXPERIENCED THE HIGHEST TRANSIT DELAYS?

SQL

```sql
SELECT
    p.port_name,
    p.country,
    p.base_dwell_days,
    ROUND(AVG(c.transit_delay_days), 2) AS
avg_transit_delay_days,
    ROUND(MAX(c.transit_delay_days), 2) AS
max_transit_delay_days,
    COUNT(c.record_id) AS total_port_calls
FROM crisis_metrics c
JOIN ports_congestion_master p
    ON c.port_id = p.port_id
GROUP BY
    p.port_name,
    p.country,
    p.base_dwell_days
ORDER BY avg_transit_delay_days DESC;
```

Average Transit Delay (Days)

KPI · AVERAGE PORT DELAY

~8 DAYS

INSIGHT

All four analyzed ports remain clustered around **8 days** of average transit delay, with Jeddah Jeddah recording the highest delay at **8.06 days.**

BUSINESS QUESTION 04

HOW DID TRADE VOLUME AND GHG EMISSIONS CHANGE ACROSS DISRUPTION PHASES?

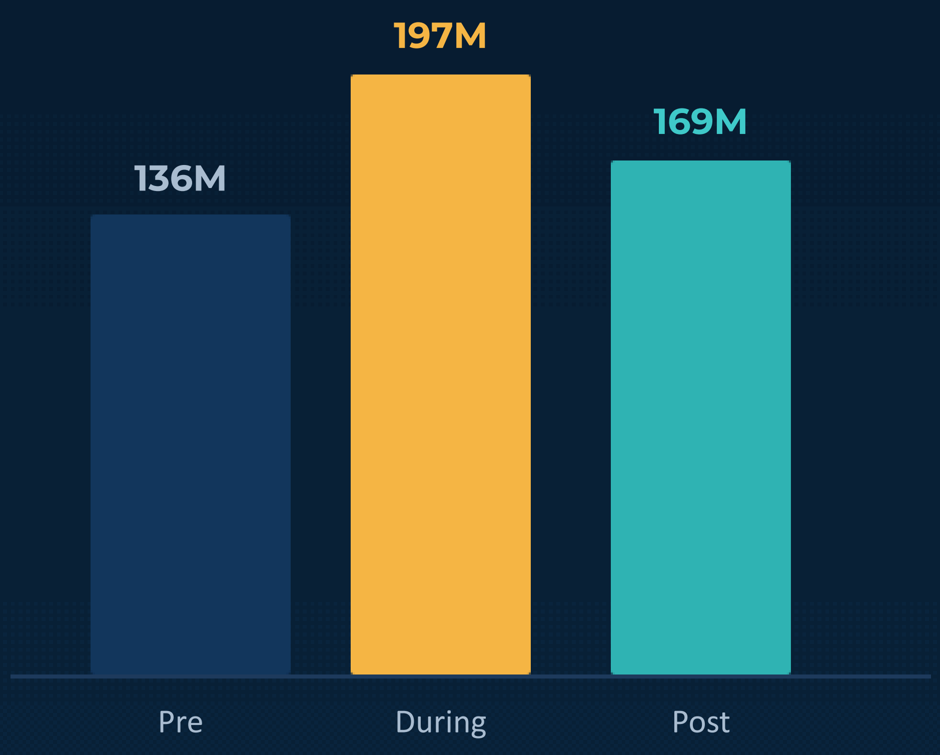
SQL

```
SELECT
    c.period_status,
    COUNT(c.record_id) AS
total_shipments,
    ROUND(SUM(c.trade_volume_tons),
2) AS total_trade_volume_tons,
    ROUND(AVG(c.trade_volume_tons),
2) AS avg_trade_volume_per_shipment,
    ROUND(SUM(c.ghg_emission_tons),
2) AS total_ghg_emission_tons
FROM crisis_metrics c
GROUP BY c.period_status
ORDER BY total_trade_volume_tons
DESC;
```

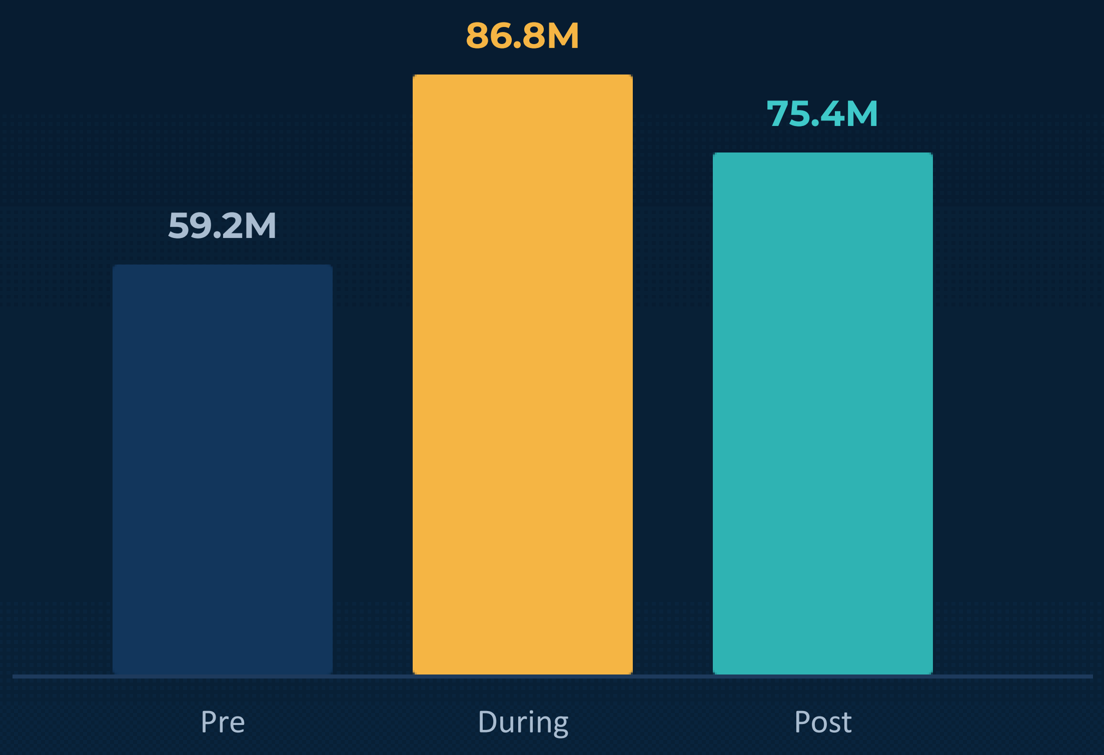
KPI · PEAK EMISSIONS

86.8M TONS GHG

Total Trade Volume (million tons)



Total GHG Emissions (tons)



Exact GHG values — Pre: 59,205,286 · During: 86,784,893 · Post: 75,362,151 tons. Note: project dataset analytical results, not UNCTAD raw shipment data.

INSIGHT

The During-Disruption phase records the highest modeled trade volume and GHG emissions, making it the most operationally intensive phase in the analytical dataset.

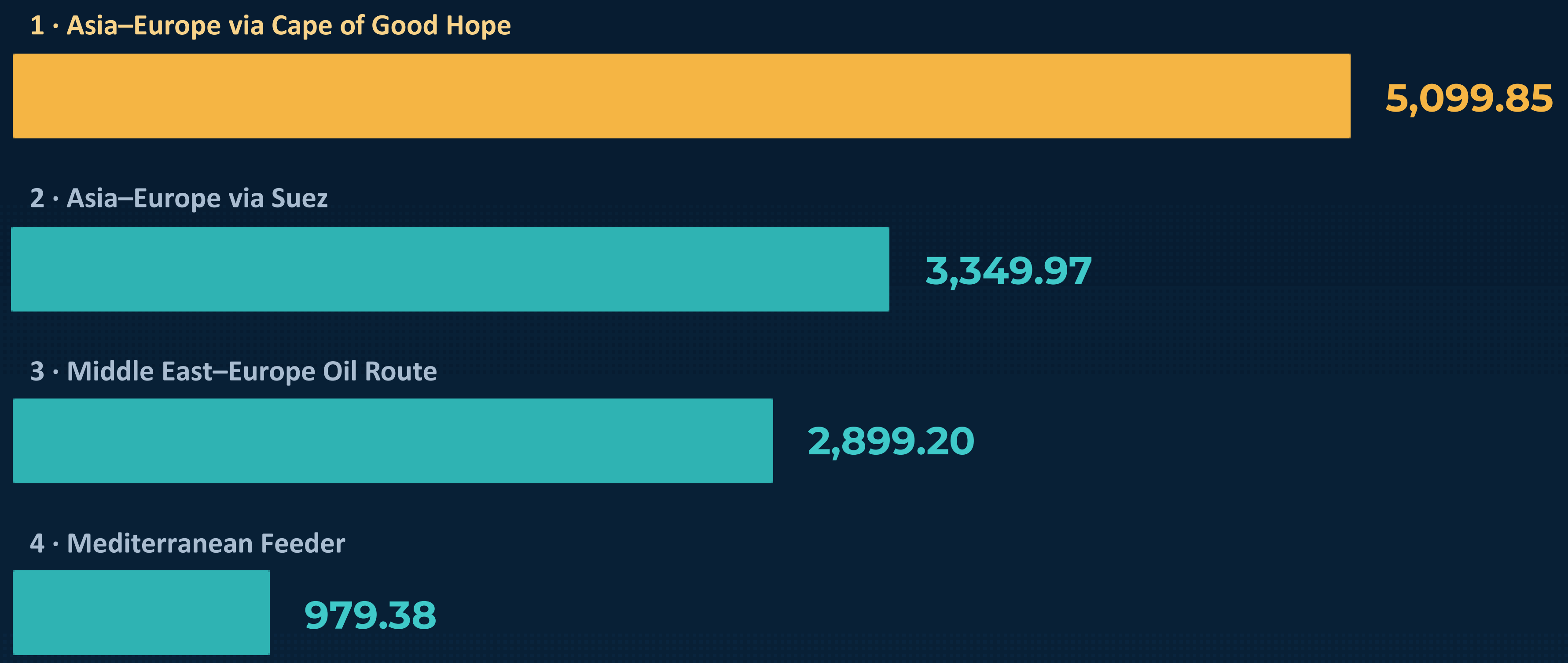
BUSINESS QUESTION 05

WHICH TRADE CORRIDOR APPEARS MOST OPERATIONALLY VULNERABLE?

SQL

```sql
WITH RouteMetrics AS (
    SELECT
        r.route_name,
        COUNT(c.record_id) AS total_shipments,
        ROUND(AVG(c.distance_nm), 2) AS avg_distance,
        ROUND(AVG(c.transit_delay_days), 2) AS
avg_delay,
        ROUND(AVG(c.freight_rate_usd), 2) AS
avg_freight_rate,
        ROUND(
            (AVG(c.distance_nm) * 0.3)
            + (AVG(c.transit_delay_days) * 50)
            + (AVG(c.freight_rate_usd) * 0.2),
            2
        ) AS composite_risk_score
    FROM crisis_metrics c
    JOIN routes_master r
        ON c.route_id = r.route_id
    GROUP BY r.route_name
)
SELECT
    route_name,
    total_shipments,
    avg_distance,
    avg_delay,
    avg_freight_rate,
    composite_risk_score,
    RANK() OVER (
        ORDER BY composite_risk_score DESC
    ) AS risk_rank
FROM RouteMetrics;
```

Composite Vulnerability Score (Ranked)



INSIGHT

The Cape of Good Hope ranks highest on the composite vulnerability score, combining the strongest modeled exposure across distance, transit delay, and freight rate.

FORMULA

Composite Score = (Avg Distance × 0.3) + (Avg Delay × 50) + (Avg Freight Rate × 0.2)

Methodology: weights and scaling are project-specific analytical assumptions.

OVERALL BUSINESS FINDINGS

WHAT DOES THE ANALYSIS TELL US?

ROUTE

+3,300 NM

Suez-dependent Asia–Asia–Europe routing carries the largest modeled diversion burden.

COST

NON-UNIFORM FREIGHT RESPONSE

Freight-rate patterns differ by vessel type and disruption phase.

DELAY

~8 DAYS

All four analyzed ports ports show ~8 days average transit delay. delay.

ENVIRONMENT

86.8M TONS GHG

During-Disruption phase records the highest aggregate modeled GHG emissions.

RISK

5,099.85

Cape of Good Hope corridor ranks highest highest on the composite vulnerability vulnerability score.

ROUTE DISRUPTION → COST & DELAY PRESSURE → ENVIRONMENTAL IMPACT → CORRIDOR VULNERABILITY

FINAL RECOMMENDATIONS

FROM SQL FINDINGS TO BUSINESS ACTION

How can supply-chain operators respond to the risks identified? *PROBLEM → FINDING → RECOMMENDATION*

PROBLEM · Large route diversion burden	FINDING · +3,300 nm Asia–Europe via Suez	RECOMMENDATION · Develop pre-planned alternative routing alternative routing and contingency corridors.
PROBLEM · Freight exposure varies by vessel type	FINDING · Freight-rate patterns patterns vary by vessel and phase	RECOMMENDATION · Monitor vessel-specific freight exposure exposure and prepare contingency contracts.
PROBLEM · High transit delays	FINDING · ~8 days average delay	RECOMMENDATION · Pre-plan alternative ports and surge surge capacity.
PROBLEM · Higher emissions during disruption	FINDING · 86.8M tons during disruption phase	RECOMMENDATION · Include emissions in route and speed speed decisions.
PROBLEM · Corridor vulnerability concentration	FINDING · Cape of Good Hope scores highest	RECOMMENDATION · Prioritise corridor monitoring, scenario scenario planning, contingency routing.

"Resilient supply chains require more than alternative routes—they require continuous visibility into distance, cost, delay and environmental exposure."

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THANK YOU

Questions & Discussion

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